eProject - Java Programmer's Guide

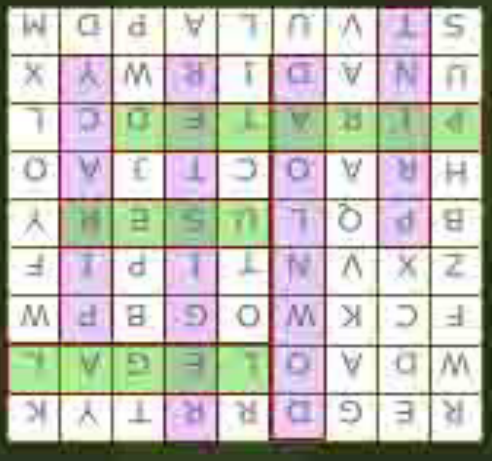
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Solution



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eProject - Java Programmer's Guide

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Preface

The eProject guide provides all the necessary information to students in developing a solution for the work process. Important functionalities required in the application are explained. Students have to understand the workflow and design the project accordingly.

The project specification is organized in the following order:

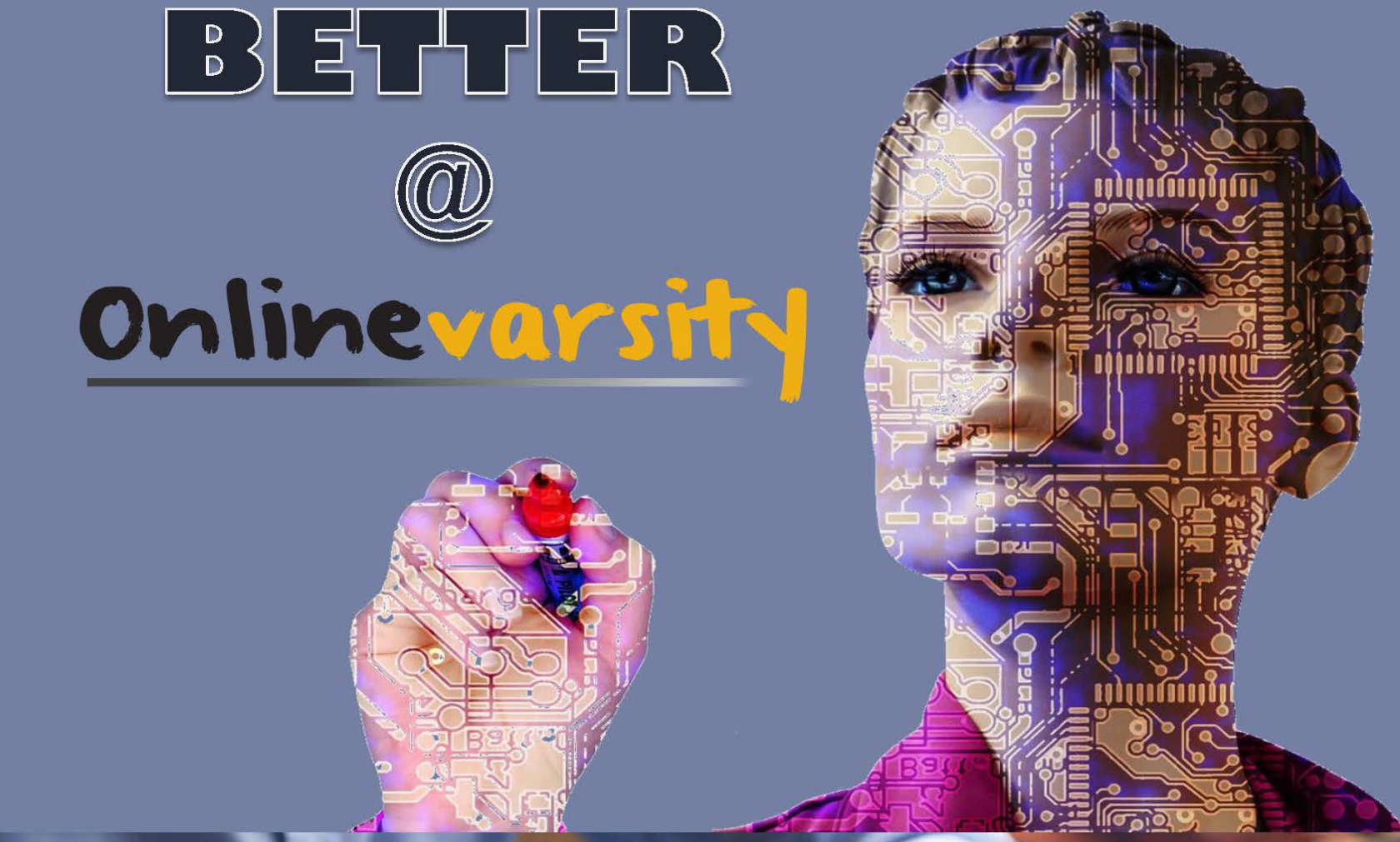
Introduction provides information related to the software industry. Process flow, suggested modules of the application and their description are also dealt with in the introduction.

Documentation describes different formats for collecting and maintaining the information. These can be used as guidelines.

Case Study provides source of information that can be used by the students while developing the project.

Sample eProject is for understanding the different aspects covered in project and the way of presenting them.

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Introduction to the Book

The best way to learn something is to apply its principles and then to test them. Similarly, the best way to evaluate the knowledge you gained is to test its application through project work. This eProject Guide, which has been prepared following the best practices in the industry, will help you to have the experience of going through a live project and will teach you the essentials of successful development of IT projects.

The eProject Guide will help you to:

- Analyze a project
- Design the specifications of the project
- Develop the solution
- Maintain disciplined documentation for the work done
- Work in groups

The ability to work in a group is a very vital quality for anybody desiring to join the software industry. Your project group will consist of 3-4 members. The Faculty will assign you to a project group and select the Leader for the Group (Group Leader).

This eProject Guide reiterates the commitment of eProjects Team in keeping up its tradition of providing innovative, career oriented professional education.

Religiously following the given systematic approaches in this book would prepare the students to get the real life experience of handling projects, as the practices listed here have been extracted from the current industry norms. Thus, such an exercise would prepare you for joining the software development industry.

Wishing you the very best.

eProjects Team

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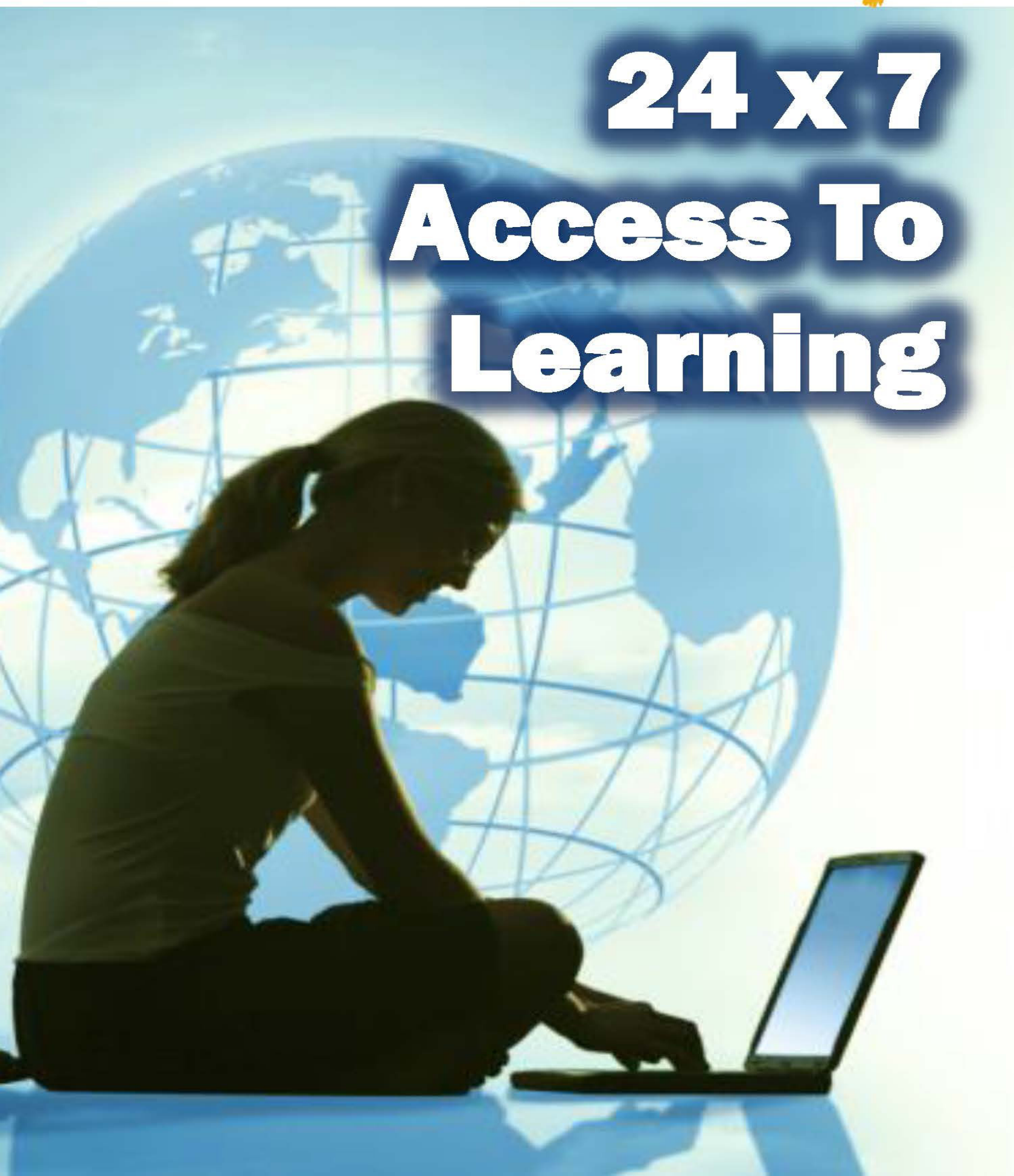
3.9 Checklist of Validations

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Annexure A: CASE STUDY

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eProject - Java Programmer's Guide

1. Introduction

1.1 Team

Implementation of any new computerized system involves a team of people. An ideal team consists of following members:

- **Project Manager**: The Project Manager coordinates the project. Apart from being knowledgeable about software, the person chosen as the Project Manager must have good writing skills. A Project Manager must have enough experience in this field so that he/she can be successful in managing the implementation. He/she is involved not only in team management, but also in activities such as resource allocation, project planning, reporting, and so forth; all of which form a part of the responsibilities.

- **Project Leader**: The Project Leader leads the project team. A Project Leader essentially decides which tasks are to be performed by each team member and how much time should be allotted to each project development phase.

- **Analyst**: The analyst studies the requirements of the system and defines the problem. The analyst determines the inputs, outputs, and processes involved in transforming those inputs into outputs. The analyst acts as the 'Technical Expert' and studies the technologies that are to be implemented to develop the project.

- **Developer/Programmer**: The developer builds the user interface according to the specifications prepared by the analyst. Next, the developer builds a prototype of the system. After receiving client approval on the prototype, the developer adds the necessary code to make the prototype a full-fledged system.

- **Tester**: The tester tests the functionality of the application. Test data is used to check if the program is able to execute without causing any errors. Test data may be live data extracted from existing records in the system or dummy data. The tester also verifies the integrated application's functionality with test data.

- **Implementation Engineer**: The Implementation Engineer ports the final product to the Client's computers. The Implementation Engineer will ensure that the installation process has been carried out accurately, and hands over the system to the client.

- **Maintenance Engineer**: The maintenance engineer is responsible for taking care of or maintaining the system that has been built. Maintenance includes extending troubleshooting support and performing software upgrades in case of changes in the external system.

1.2 Tips for a Good Application

The seven basic steps for creating an effective application are as follows:

1. Define your target audience
2. Organize your concepts, information, and material
3. Create a directory structure
4. Implement storyboarding – prepare a sketch of contents, sequence, and layout of the forms you intend to create
5. Build a prototype of your application
6. Test the prototype and implement the required changes
7. Deploy the application on the server, if applicable

Before designing an application, you must first conceptualize the application. The goals and the objectives must be clearly defined. Think about what the application goals and missions may be to inform, promote, educate, research and report, or to simply entertain. These goals must be defined and should be in line with the company's values and mission.

Annexure A shows a case study of a project.

1.3 Project Objectives

1. Develop a desktop application based on the problem specifications given
2. Design a professional-looking Graphical User Interface (GUI) for the application
3. Integrate all the modules to form a complete solution

1.4 Project Deliverables

Following are the deliverables that have to be submitted on the completion of the project:

1. Complete Application
2. User Manual
3. Installation Guide

1.5 Project Conduct

During project development, three meetings would be held within the team.

➢ **First Meeting**



In this meeting, students need to discuss the eProject Specification. They need to work upon:

- Problem Statement
- Table/File Structure
- Program/Code Specifications
- Reports to be generated

➢ **Review of First Meeting**

After the first meeting, students will review and discuss the details of the meeting. They will check if the structure is correct and the forms are designed as per the design specifications.

- **Second Meeting**



In this meeting, students will discuss the design. They need to consider the following:

- Understanding of the problem
- Design of the forms
- Validations required

- **Review of Second Meeting**

After the second meeting, students will review and discuss the details of the meeting. They will check whether the design of forms are correct and proper verification and validation rules are being implemented.

- **Third Meeting**

In this meeting, students will discuss and present the final application. They will consider the following:

- Implementing all the specifications mentioned in the previous meetings
- Form Design
- All the required validations

- Integrating all the project modules such as the forms, reports, and so forth
- User Manual and Installation Guide



2. Steps for a Great Project

- Analyze and understand the user requirements
- Do the detailed analysis for the project
- Identify the resources available
- Plan and maintain a schedule for the various activities to be done in the project
- Check and test all modules
- Document each and every detail

Documentation Section

In this section, we will look into various formats used for documenting the project.

This section contains different formats, which can also be referred to, in Annexure A, in the case study.

Students can use their own additional formats in the project documentation.

3. Documentation Section

This format can be used as the first page of your project report, duly signed by the Center Academics Head.

This is to certify that

Mr./ Ms _____

Has successfully Designed & Developed

_____

Submitted by:

_____

Date of Issue:_____

Authorized Signature:_____

3.1 Problem Definition

Problem Definition:

In this form, you can define the problem statement. You can design a pointwise list of the requirements as mentioned in the eProject Specification.

Form No.1/eProjects/PS/Ver 1.0

3.2 Customer Requirement Specification (CRS)

Client:

Business/Project Objective:

(can address organization/business overview, products, concerns, and expectations from the system)

Inputs provided by the Client:

- Inputs to the System
- Outputs from the System
- Process Involved in the System
- Expected Delivery Dates
- List of Deliverables

Hardware Requirements:

Software Requirements:

Scope of the Work (in brief):

Form No.2/eProjects/CRS/Ver 1.0/1 of 1

3.3 Architecture and Design of the Project

You can provide the Architecture and Design for your eProject in this format.
You can refer to Annexure A Case Study to understand the use of Architecture and Design of the Project.

Form No. 3/eProjects/Design/Ver 1.0/1 of 4

3.4 Flow Chart

A flow chart is a graphic representation of an algorithm or a process.

You can provide the flow charts for your eProject in this format.
You can refer to Annexure A Case Study to understand the use of flow charts.

Form No. 3/eProjects/Design/Ver 1.0/2 of 4

3.5 Data Flow Diagram

A Data Flow Diagram (DFD) is a graphical representation of the flow of data.

You can provide the Data Flow Diagrams for your eProject in this format.
You can refer to Annexure A Case Study to understand the use of Data Flow Diagrams.

Form No. 3/eProjects/Design/Ver 1.0/3 of 4

3.6 Entity Relationship (ER) Diagram

Entity Relationship is used to describe information needs and/or the type of information that is to be stored in the database.

You can provide the Entity Relationship Diagrams for your eProject in this format.

Form No. 3/eProjects/Design/Ver 1.0/4 of 4

3.7 Database Design/Structure

Table is a set of data elements (cells) that is organized, defined, and stored.

Using this format, you can provide the Table Structure for your eProject in this format.
You can refer to Annexure A Case Study to understand the use of Table Structures.

Form No. 4/eProjects/DD/Ver 1.0/1 of 1

3.8 Task Sheet

Project Ref. No.:		Project Title:	Activity Plan Prepared By:	Date of Preparation of Activity Plan:			
Sr.No.	Task			Actual Start Date	Actual Days	Team Mate Names	Status

Form No.5/eProjects/TS/Ver 1.0/1 of 1

3.9 Checklist of Validations

Option	Validated
Do all numeric variables have a default value of zero?	
Does the administrator have all the rights to create and delete the records?	
Are all the records properly fed into the appropriate database?	
Have all the modules been properly integrated and are completely functional?	
Have all the Design and Coding Standards been followed and implemented?	
Is the GUI design consistent all over?	
Is the navigation sequence correct through all the forms/ screens in the application?	
Is exception handling mechanism implemented in all the screens?	
Are all the program codes working?	

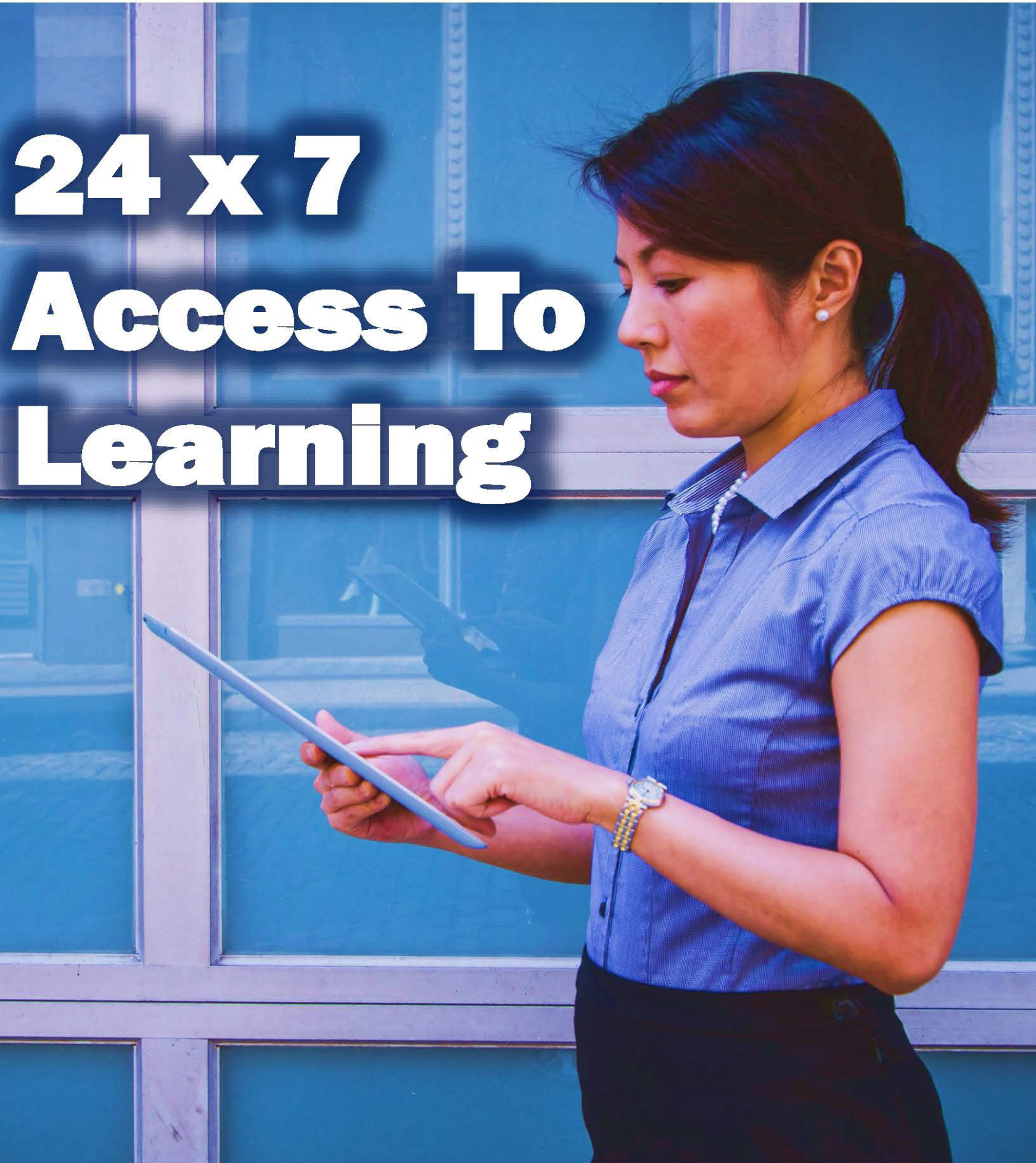
Form No. 6/eProjects/VAL/Ver 1.0/1 of 1

3.10 Submission Checklist

Sr. No.	Particulars	Yes	No	NA	Comments
1.	Are all the users able to search for a particular record?	Yes			
2.	Are all the old records properly saved and retrieved when required?	Yes			
3.	Have all the modules been properly integrated and are completely functional?	Yes			
4.	Are the GUI contents devoid of spelling mistakes?	Yes			
5.	Is the application user-friendly?	Yes			
6.	Is the project published properly into a setup file?	Yes			

Form No. 7/eProjects/CKL/Ver 1.0/1 of 1

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Annexure A

With the rise in retail business, retailers need to maintain records of every product, whether it is being bought or sold. In addition, their operations have to be efficient and should be documented properly. Many organizations nowadays are deploying industry standard quality management methodologies based on ISO 9001 and/or any other quality standards relevant to their type of industry. There is a need to offer a software solution to offset cost by substituting paper based expenses with increased control, efficiency, productivity, and live documentation.

This software solution will enable to add, modify, and view customer details and product details. It will also help to perform product sales and transactions. Also, the software should be capable of storage, retrieval, and manipulation of data in appropriate database tables.

End users of this software solution are retailers and their staff.

Details of the software solution or application are as follows:

- An 'Admin' is the person with rights and permissions to all tasks in the software application system. Admin can register himself/herself with the application. This registration will enable him/her to log in to the system and use the application.

- An Admin table in the database will store the login details of the Admin person such as Login ID, Name, Organization ID, and Password.

- 'Admin' user should login before using the application. The 'Admin' user would input Login ID and Password which would be matched against the stored Login ID and Password.

- Admin user should enter Product and Category details into respective tables. The application should be capable of displaying products belonging to a specific category, if the Category ID is provided.

- Customers are people who actually buy products in the store. However, they would not be directly using the application. Retailer staff such as the Admin will perform tasks in the application on behalf of customers. Before doing any transaction, the customer details should be present in the application. Retailer staff is responsible for getting all the customer details and adding it to the customer table. Only an 'Admin/Retailer' user would be able to register a customer.

- The application should auto-generate a customer id for uniquely identifying the customer. Before saving customer details, the application should check if the customer record already exists.

- Once registered, the customer details can be used to perform transactions through the 'Sales' module. The Sales module will also use Product and Category details.

- Whenever a sale of products is made to a customer, the data should be updated in the database

so that respective notifications can be shown for current outstanding balance and product quantity.

- All the product details should be stored in a Product Details database table. Product details can include Product Id, Product Name, Description, Product Quantity, and Price Per Unit.

- The application should display the current price of selected products that the customer wants to buy and should accept quantity for these selected products. The application should then compute and display total price under each category as well as grand total price. For example, if the customer purchases four items of a product that costs 5 dollars each, the price will be 20 dollars, and then if the customer further buys two items of 20 dollars each, then the grand total price will be 60 dollars. This has to be computed and displayed by the application.

- Upon successful sale, Customer ID, Product Name, Amount, Price per Unit, Quantity of Product, Date of Transaction, and so on would be recorded in the appropriate transaction table such as Product Sales.

- Based on sales, the application should update the value of Product quantity in the Product master table.

- 'Admin' user cannot sell products for the customer in the system if:

    - Quantity of products that the customer wants to buy is less than the Quantity of products that is present in the Products table.

Proposed Solution

The proposed solution of the project is a desktop application named Regal Soft.

The entry point will be a login screen which is responsible for validating the users. The application will allow the Admin to log in to the system with specified Login ID and Password. After logging in, Admin can add/update Customer details and Product details. Admin will also be responsible for managing sales details.

Database tables will be used to store information related to Products, Customers, and Sales info.

Regal Soft is a retail management software system that brings together various tasks related to retail into a single, centrally-managed solution. This software solution is an end-to-end suite designed especially for multi-channel retailers that will enable enhanced customer service across channels, while driving growth and increased revenue.

Project Specifications

The application is to be developed for Windows platform using Java SE 8, NetBeans 8, and SQL Server 2012/2016.

The application should perform the following functions:

- Store information for managing products sales and purchase in database tables
- Register a new Admin user
- Allow login for registered Admin
- Following options should be available only to the Admin
  - Products management
  - Customer information management
  - Generate bill
- Register a customer
- Search for a particular customer's information
- View detailed customer information such as:
  - Customer information
    - Customer ID, Customer Name, Gender, Phone, Address, Email, Product Sales for this customer, and Customer's balance (Outstanding)
  - Customer purchase list
    - Customer ID, Product name, Category, Amount, Price, and Quantity
- Update customer information
- Perform sales of products
- View products information

Hints for developing the Solution:

Create a desktop application that would have menu options and screens (wherever applicable) for all these tasks that should be performed by the application. Implement the logic and functionality.

ACKNOWLEDGEMENT

I would like to acknowledge all those who have given moral support and helped me make the project a success.

I wish to express my gratitude to the eProjects Team at the Head Office, who guided and helped me. I would also like to express my gratitude to all the staff members of my center for not only providing me with the opportunity to work with them on this project, but also for their support and encouragement throughout the process.

I also express my sincere gratitude to the my project guide at the organization, for his/her valuable guidance and support for the completion of this project.

And finally, I would like to offer many thanks to all my colleagues for their valuable suggestions and constructive feedback.

Problem Definition

After reading the project specifications, the developer states the scope of the project very briefly.

The **Problem definition** in the Regal Soft application would be: With the rise in retail business, retailers need to maintain records of every product, whether it is being bought or sold. In addition, their operations have to be efficient and should be documented properly. Many organizations nowadays are deploying industry standard quality management methodologies based on ISO 9001 and/or any other quality standards relevant to their type of industry. There is a need to offer a software solution to offset cost by substituting paper based expenses with increased control, efficiency, productivity, and live documentation.

This software solution will enable to add, modify, and view customer details and product details. It will also help to perform product sales and transactions. Also, the software should be capable of storage, retrieval, and manipulation of data in appropriate database tables.

End users of this system will be retailers and their staff.

Form No.1/eProjects/PS/Ver 1.0

Customer Requirement Specification (CRS)

Client: Regal Soft application

Business/Project Objective:

(can address organization/business overview, products, concerns, and expectation from the system)

Develop a retail management software system that brings together various retail operations and tasks into a single, centrally-managed solution. This software solution is an end-to-end suite designed especially for multi-channel retailers that will enable enhanced customer service across channels, while driving growth and increased revenue.

Inputs provided by the client:

- Inputs to the existing system
- Outputs from the existing system
- Process involved in the application
- Expected delivery dates
- List of deliverables

Hardware Requirements:

- Intel Core i3 Processor or higher
- 4 GB RAM or above
- Color SVGA
- 120 GB Hard Disk space

Software Requirements:

- Microsoft Windows 7 or Microsoft Windows 8.1 or higher
- Java SE 8, NetBeans 8, and SQL Server 2012/2016.

Form No.2/eProjects/CRS/Ver 1.0/1 of 2

Scope of the Work (in brief):

Depending on the decision taken by the management regarding the Regal Soft application, following are the requirements based on which the software needs to be developed:

The application should perform the following functions:

- Store information for managing the products sales and purchase in database tables
- Register a new Admin user
- Allow login for registered Admin
- Following options should be available to the Admin
  - Products management
  - Customer information management
  - Generate bill
- Register a customer
- Search for a particular customer information
- View detailed customer information. The information should include:
  - Customer information:
    - Customer ID, Customer Name, Gender, Phone, Address, Email, Product Sales for this customer, Customer's balance (Outstanding)
  - Customer Purchase list
    - Customer ID, Product name, Category, Amount, Price, Quantity
- Update customer information
- Perform sales of products
- View products information

Form No.2/eProjects/CRS/Ver 1.0/2 of 2

Architecture and Design of the System

Figure 1.1 displays the application architecture.

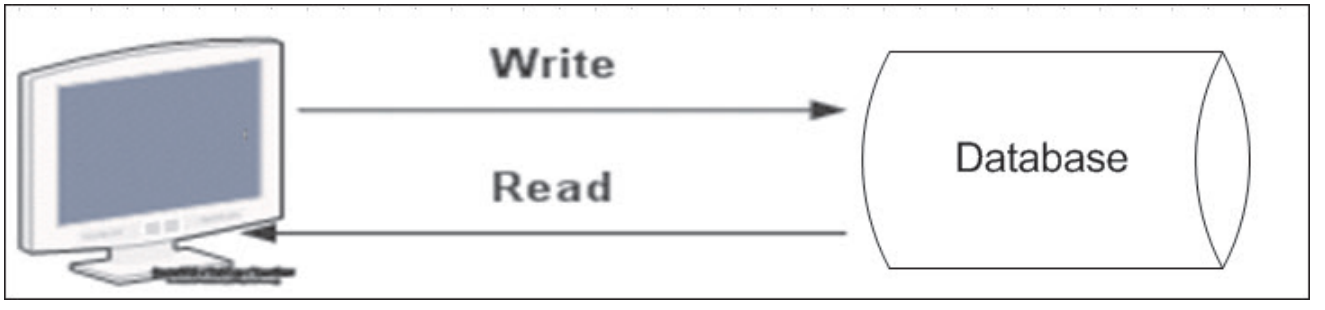


Figure 1.1: Application Architecture

Form No. 3/eProjects/Design/Ver 1.0/1 of 9

Data Flow Diagram

Figure 1.2 displays the context level diagram of the application.

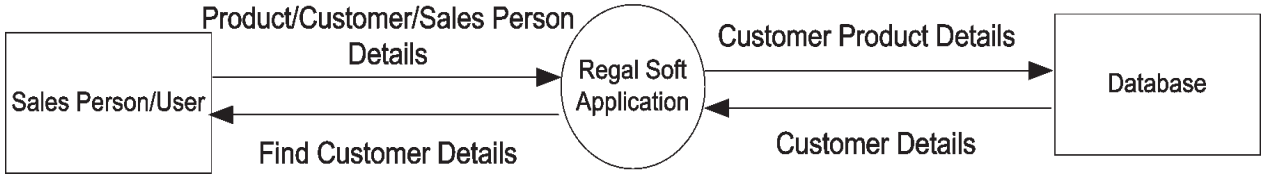


Figure 1.2: Context Level Diagram

Figure 1.3 displays the first level of Data Flow Diagram.

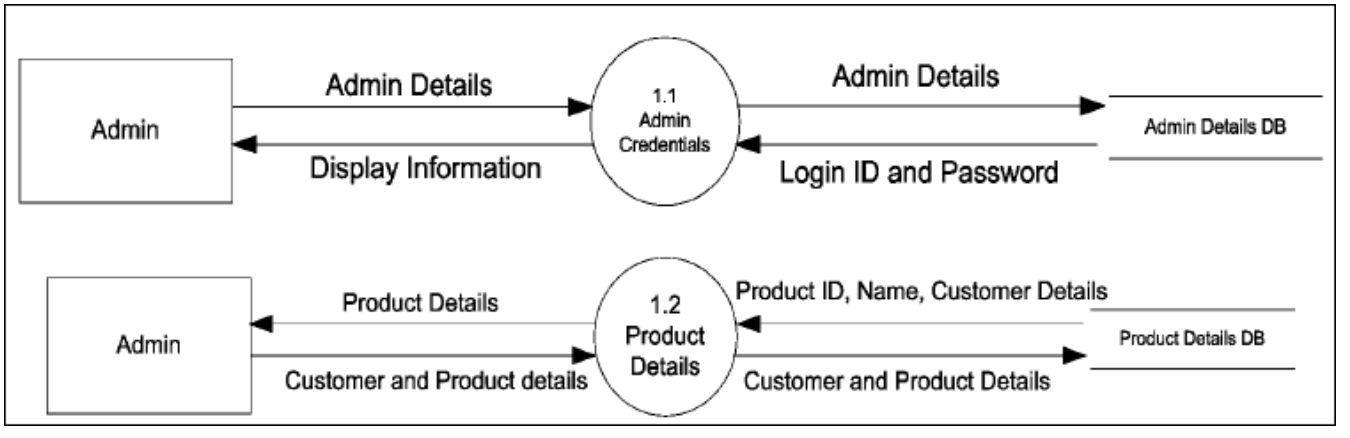


Figure 1.3: Admin and Products First Level DFD

Figure 1.4 displays the first level of data flow diagram of Product Transaction.

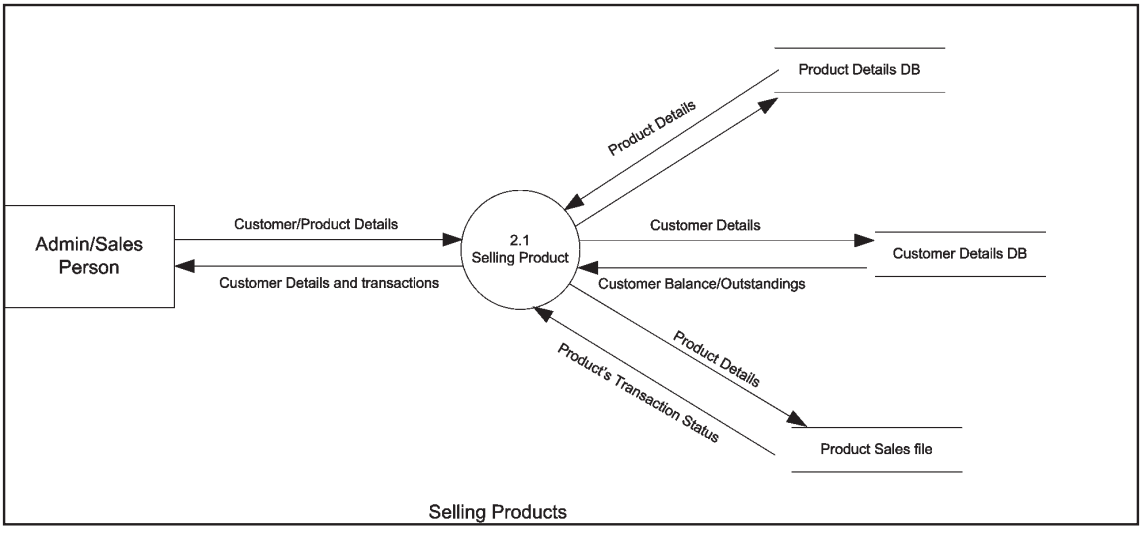


Figure 1.4: Selling Products

Form No. 3/eProjects/Design/Ver 1.0/3 of 9

Use Case

Figure 1.5 shows the use case of the application.

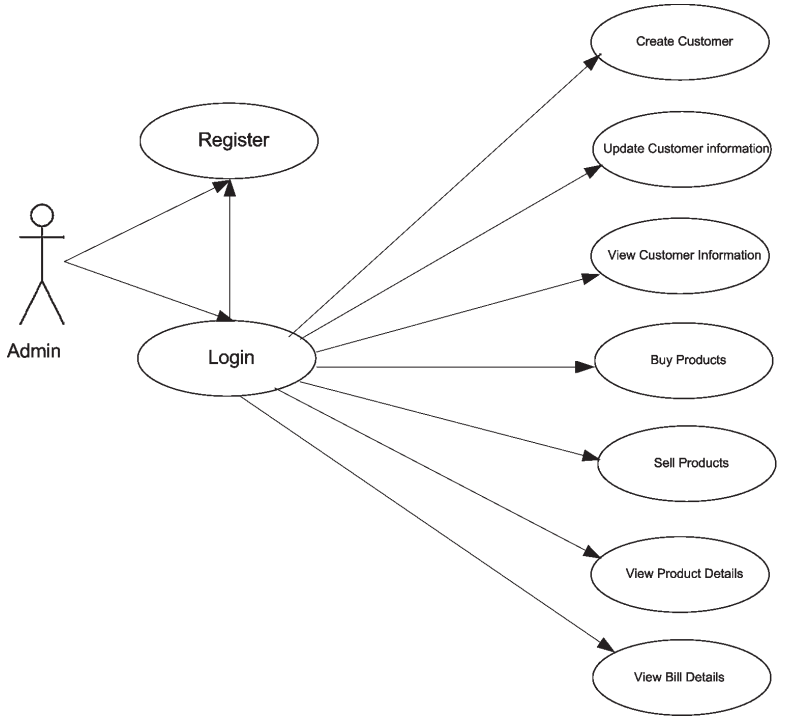


Figure 1.5: Use Case

Login

The application flow for login screen is shown in figure 1.6.

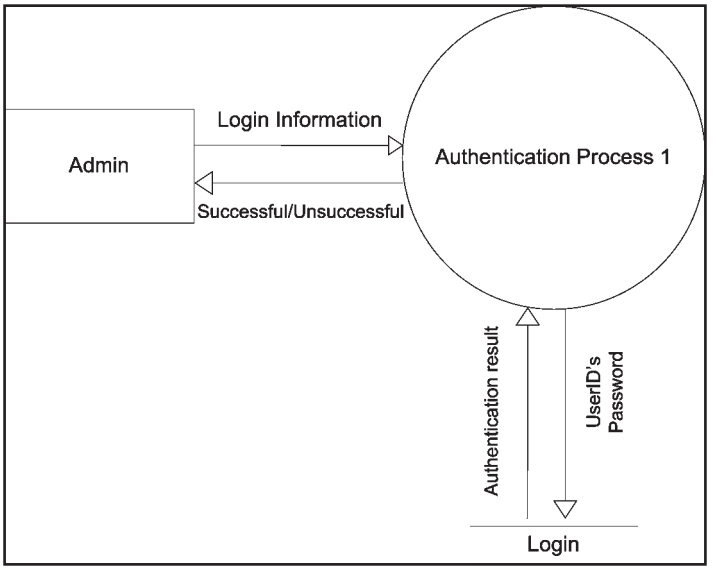


Figure 1.6: Login

Application Screen

Figure 1.7 shows the different activities associated with the application.

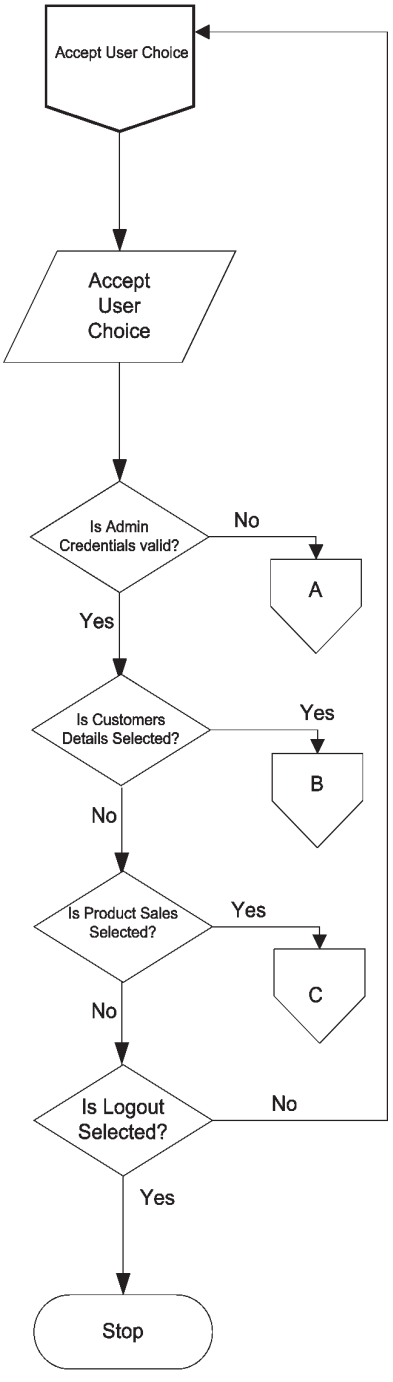


Figure 1.7: Application Modules

Form No. 3/eProjects/Design/Ver 1.0/6 of 9

Admin Process

Figure 1.8 shows the different Admin activities associated with the application.

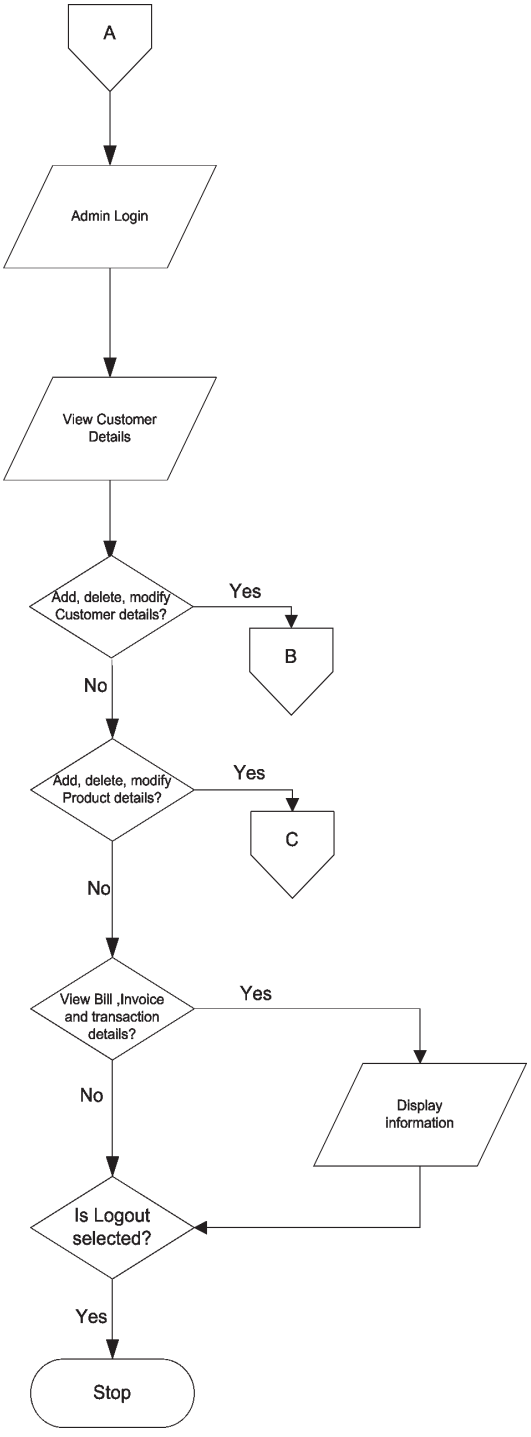


Figure 1.8: Admin Process

Form No. 3/eProjects/Design/Ver 1.0/7 of 9

Selling Products

Figure 1.9 shows the application flow for selling products.

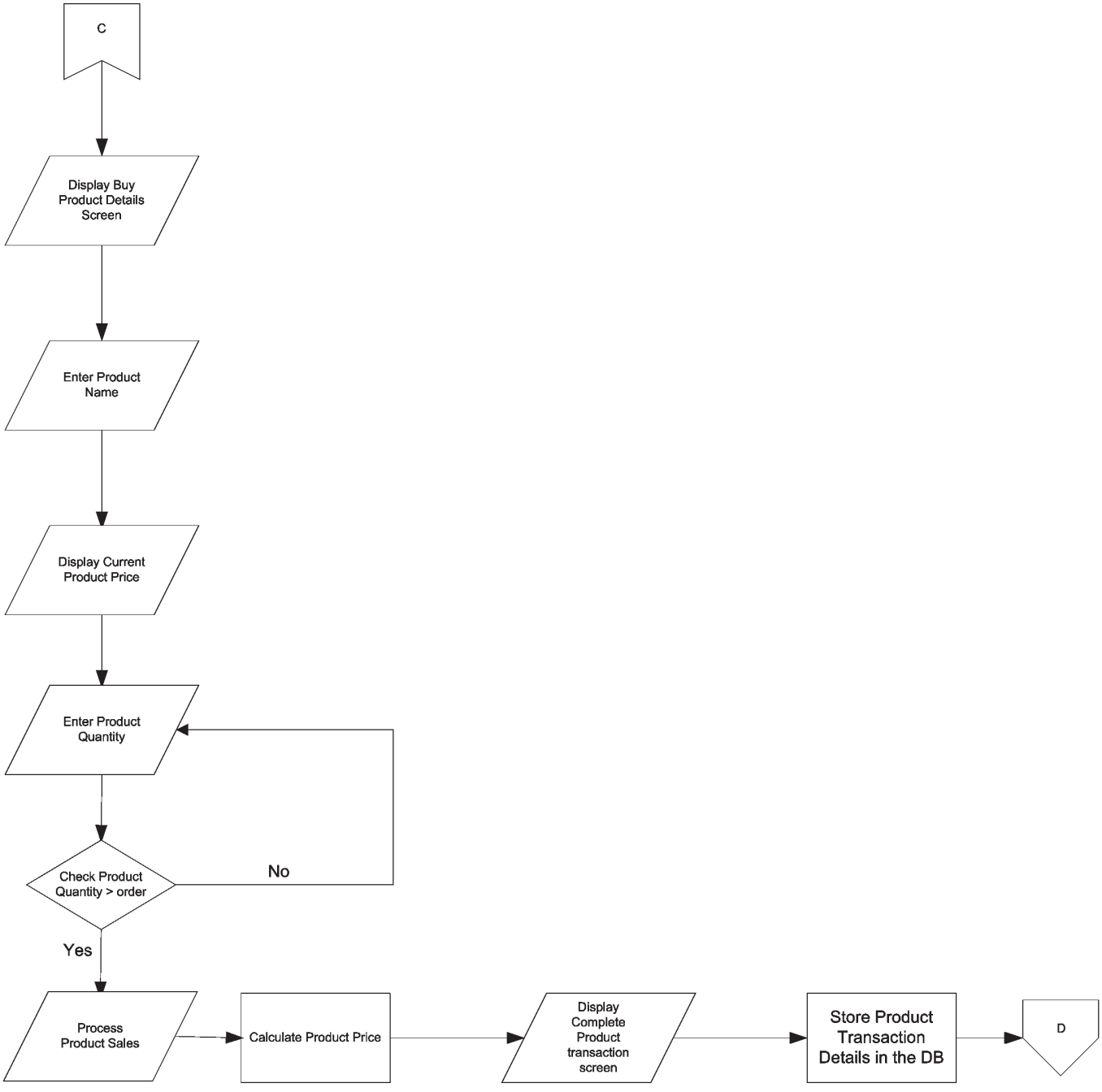


Figure 1.9: Selling Products

Form No. 3/eProjects/Design/Ver 1.0/8 of 9

Figure 1.10 shows the application flow for addition of customers.

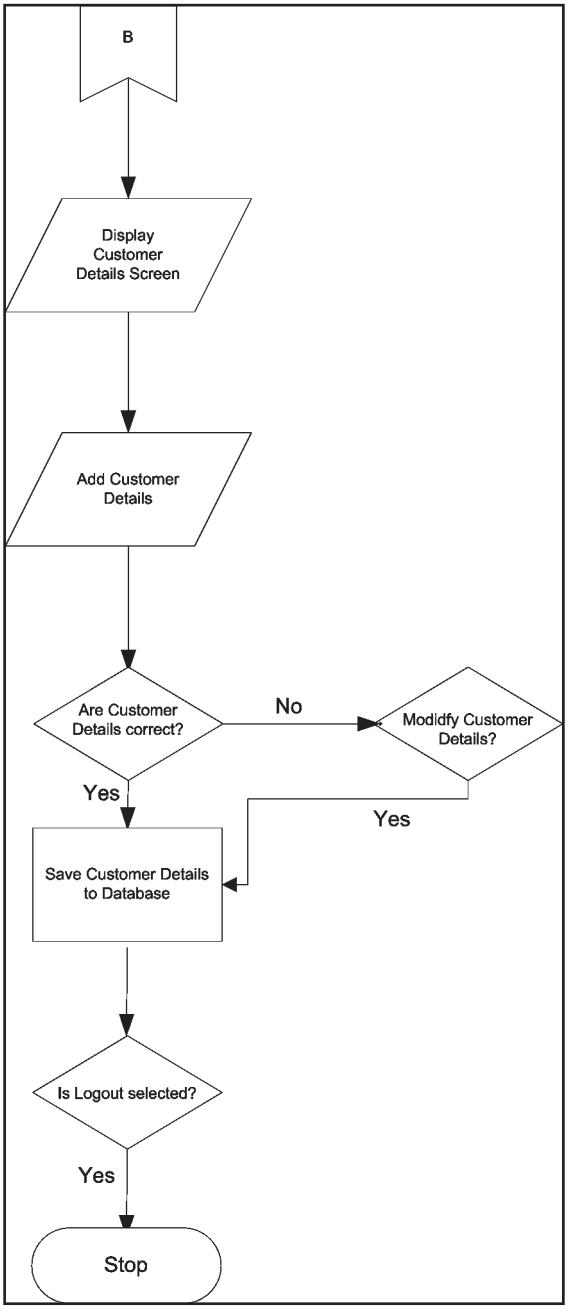


Figure 1.10: Adding Customer

Form No. 3/eProjects/Design/Ver 1.0/9 of 9

Database Design/Structure

Database structure of the different Tables is given in the following sections:

- **Customer Details**

  Customer Details table stores the customer details.

Field Name	Data Type	PK/FK	Description
Customer_ID	Varchar(50)	PK	Generates Customer ID
Customer_Name	Varchar(50)		Stores Customer Name for given ID
Customer_ Address	Varchar(100)		Stores Customer Address
Customer_mb_no	int		Stores Customer Mobile Number
Email_id	Varchar(50)		Stores email id for mentioned Customer ID
Credit_limit	int		Stores Credit limit for mentioned Customer ID
Total_balance	int		Stores total balance of the Customer
Outstanding_Balance	int		Stores Outstanding balance of the Customer

- **Admin**

  Admin table stores the details of the Admin of the retail store.

Field Name	Data Type	PK/FK	Description
User ID	Varchar(50)	Pk	Stores User ID, (unique)
Username	Varchar(50)		Stores Username associated with User ID
Password	Varchar(50)		Stores Password associated with User ID

- **Product Master**

Product Master table stores product details.

Field Name	Data_type	PK/FK	Description
Product_ID	Varchar(50)	PK	Generates Product ID
Product_Category	Varchar(50)		Stores Product Category
Product_Name	Varchar(50)		Stores Product Name
Product_Quantity	int		Stores Product Quantity
Product_Price	int		Stores Product price

- **Product Sales**

Product Sales table stores the transaction details of the product sales of all the customers.

Field Name	Data_type	PK/FK	Description
Customer_ID	Varchar(50)	FK	Stores Customer ID
Product_ID	Varchar(50)	PK	Stores Product ID
Product_Quantity	int		Stores Product Quantity
Product_Price	int		Stores Product price
Total Price	int		Stores Total price for the given customer
Date_Of_Transaction	Date		Stores Date and time for mentioned order
Payment_Mode	Varchar(50)		Stores type of payment made by the customer
Payment_made	int		Stores Payment value of the given Customer ID
Outstanding_balance	int		Stores outstanding balance of the provided customer

Task Sheet

Project Ref. No.: **Regal Soft application**		**Project Title:**	**Activity Plan Prepared By:**	**Date of Preparation of Activity Plan:**			
Sr.No.	**Task**			**Actual Start Date**	**Actual Days**	**Team Mate Names**	**Status**
1	Login Form design and coding	Regal Soft Application	Peter	1-Aug-2016	1	Mark	Completed

Form No.5/eProjects/TS/Ver 1.0/1 of 1

Checklist of Validations

Option	Validated
Do all numeric variables have a default value of zero?	Yes
Does the administrator have all the rights to create and delete the records?	Yes
Are all the records properly fed into the appropriate database?	Yes
Have all the modules been properly integrated and are completely functional?	Yes
Have all the design and coding standards been followed and implemented?	Yes
Is the design consistent all over?	Yes
Is the navigation sequence correct through all the screens in the application?	Yes
Is exception handling mechanism implemented in all the screens?	Yes
Are all the program codes working?	Yes

Form No. 6/eProjects/VAL/Ver 1.0/1 of 1

Submission Checklist

Sr. No.	Particulars	Yes	No	NA	Comments
1.	Have all the modules been properly integrated and are they completely functional?	Yes			
2.	Does each unit meet its objective and purpose? Are all the validations happening as specified in Process Design?	Yes			
3.	Have all Design and Coding standards been followed and implemented?	Yes			
4.	Is the GUI design consistent all over?	Yes			
5.	Are the codes working as per the specification?	Yes			All the modules are properly tested.
6.	Does the application's functionality resolve the client problem, and satisfy his/her needs completely?	Yes			
7.	Have the hardware and software been correctly chosen?	Yes			
8.	Additional features and utilities that give value addition to the entire project.	Yes			

Form No. 7/eProjects/CKL/Ver 1.0/1 of 1

4. User Guide

A. **System Requirements:**

No.	Items	Description
1	Operating System	Microsoft Windows 7 or Microsoft Windows 8.1 or higher
2	Database	SQL Server 2012/2016
3	Software	Java SE 8 and NetBeans 8

B. **Install and Run Application:**

Step 1: Install SQL Server 2012/2016 on the local system.

Step 2: Create appropriate database and tables as per the specifications given.

Step 3: Execute the application.

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